University of Kent

End-of-term assignment

[Data Analysis Report]

Course code: BUSB7028

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Contents

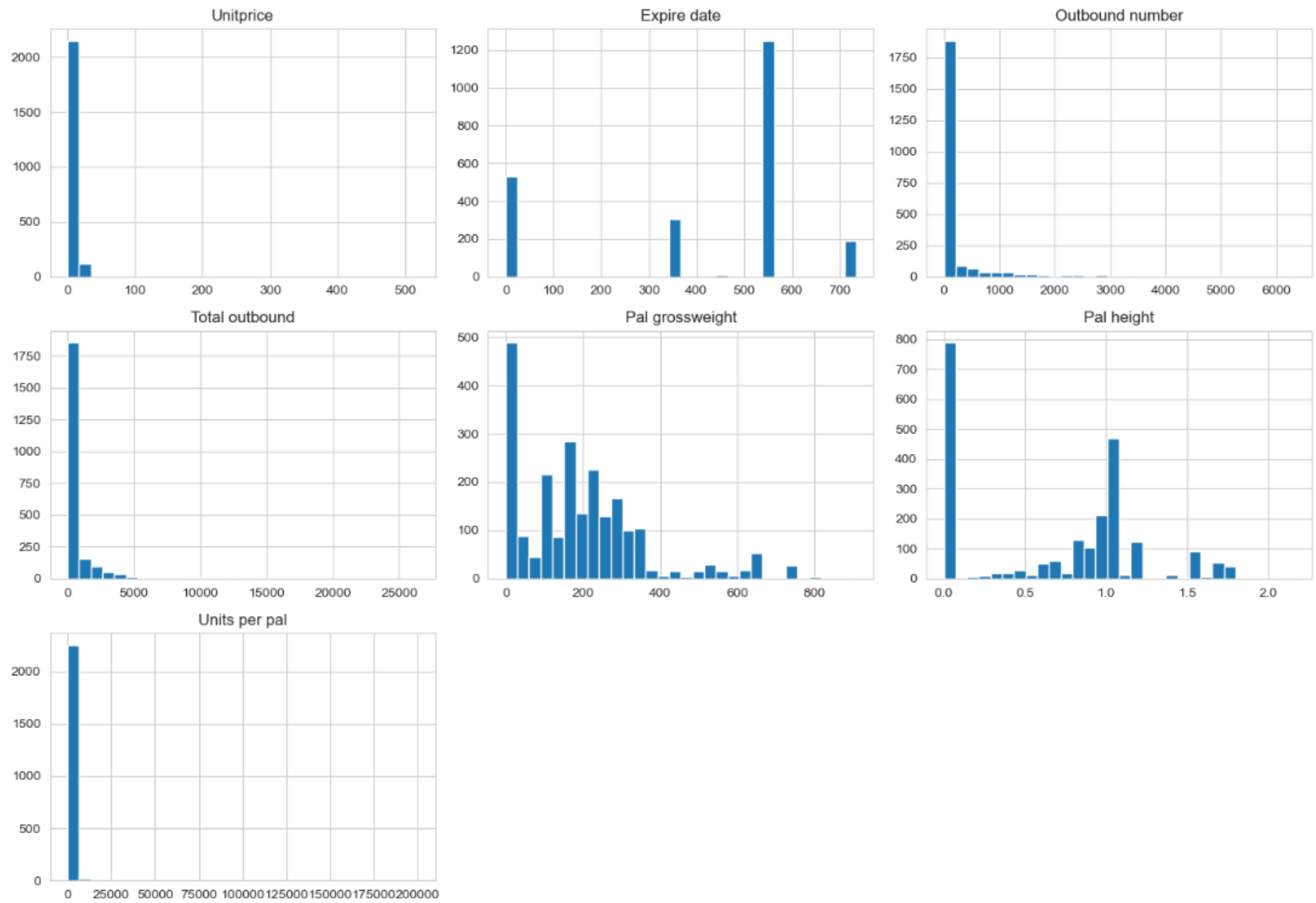
PART A

Question 1

The code starts with EDA on **sku_data.csv** by checking shape, types, missing values, duplicates, summary statistics, zeros, distributions, boxplots and correlations. The data contains 2,279 rows and 7 numeric variables, with no missing values, 357 duplicates and many zeros, so many variables are strongly skewed (see Figure A1).

Figure A1

Distribution of numeric features

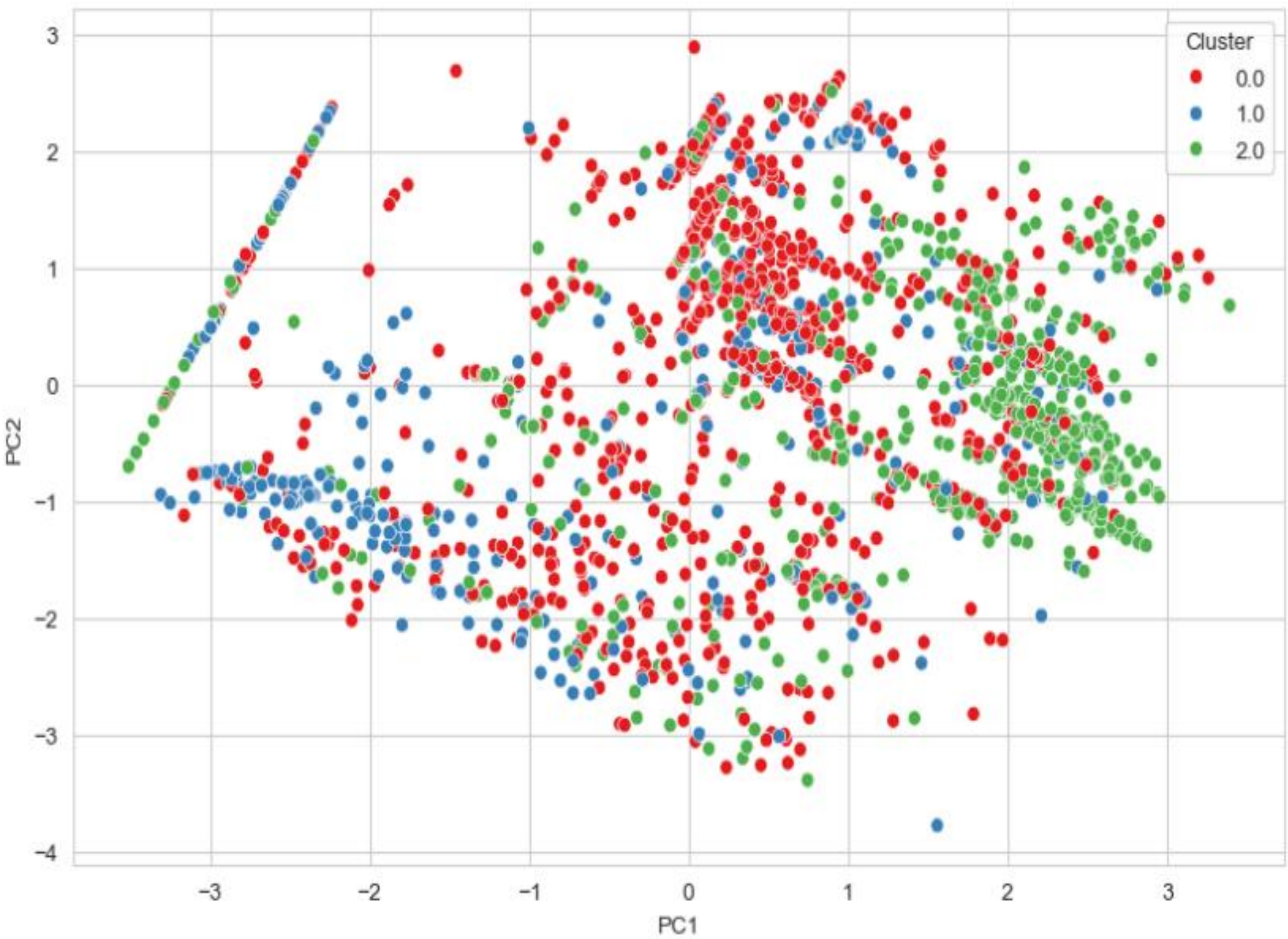


Source: author's analysis

Because **K-means** depends on distance, the code applies **log1p** and then standardises the features so large-scale variables do not dominate. It tests k = 2 to k = 10 using the elbow method and silhouette scores, then selects k = 3 for a simple final solution. The three clusters contain 1070, 496 and 713 **SKUs**. Cluster 2 is the high-demand segment, with much larger outbound number and total outbound than the others. Cluster 1 contains small, light and low-movement items, while cluster 0 is a medium group with higher price and pallet size than Cluster 1 but much lower demand than Cluster 2.

Figure A2

Two-dimensional PCA visualisation of 3 clusters



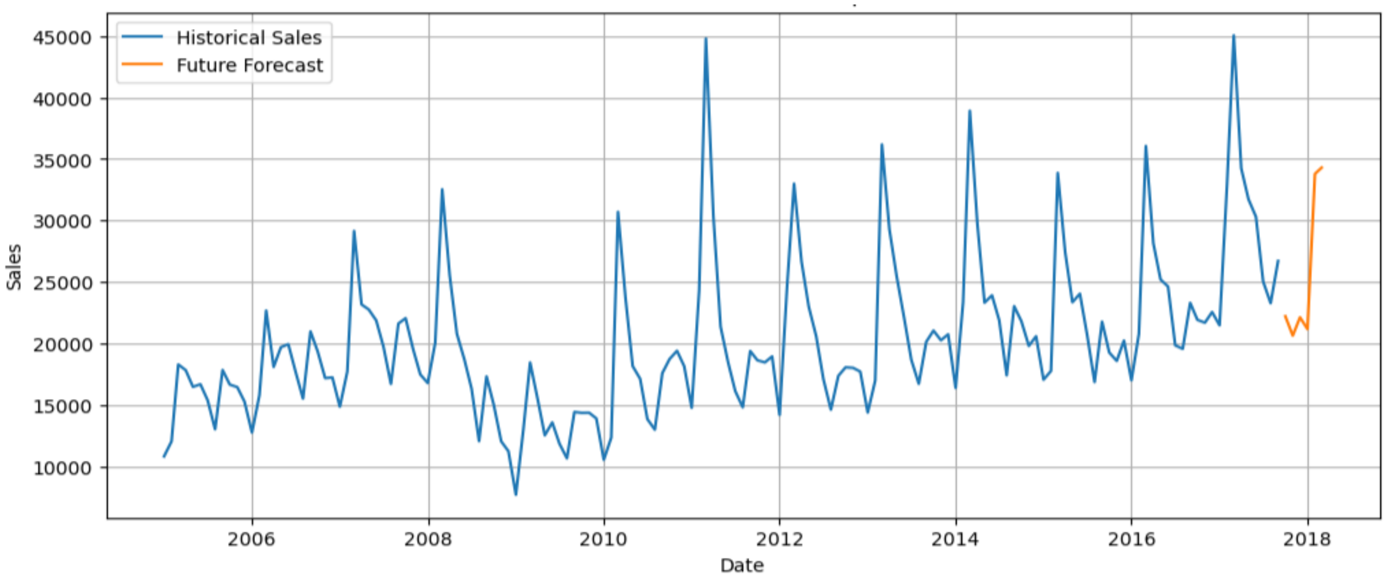
Source: author's analysis

Question 2

The time-series code cleans the compact crane sales data by forward-filling missing Year values, creating a monthly Date index and converting Sales from text into integers. It then plots the series with 3-month and 12-month moving averages to inspect trend and seasonality. For forecasting, the notebook uses **RandomForestRegressor**. Lag variables for 1, 2, 3, 6 and 12 months, rolling means and calendar features such as month and quarter are created to capture temporal patterns. The data is split chronologically, with 80% for training and 20% for testing. The model achieves $RMSE$ = 3636.89 and MAE = 2372.38, which indicates a reasonable forecast for monthly sales. Finally, the model generates a recursive 6-month forecast for October 2017 to March 2018, with expected sales roughly between 20000 and 34000 units and a stronger outlook in early 2018 (see Figure A3).

Figure A3

Future sales forecast for compact crane



Source: author’s analysis

Question 3

regression notebook first explores regression dataset by EDA again, then converts qualitative variables into dummy variables using one-hot encoding, which is appropriate for linear regressions. One category from each variable is dropped to avoid the dummy-variable trap and the duplicate binary **Job_type** indicators are removed because the Job dummies already capture employment type, with fixed as the baseline category. VIF values are low, so multicollinearity is not a major issue (see Table A1).

Table A1

VIF values for independent variables in regression model

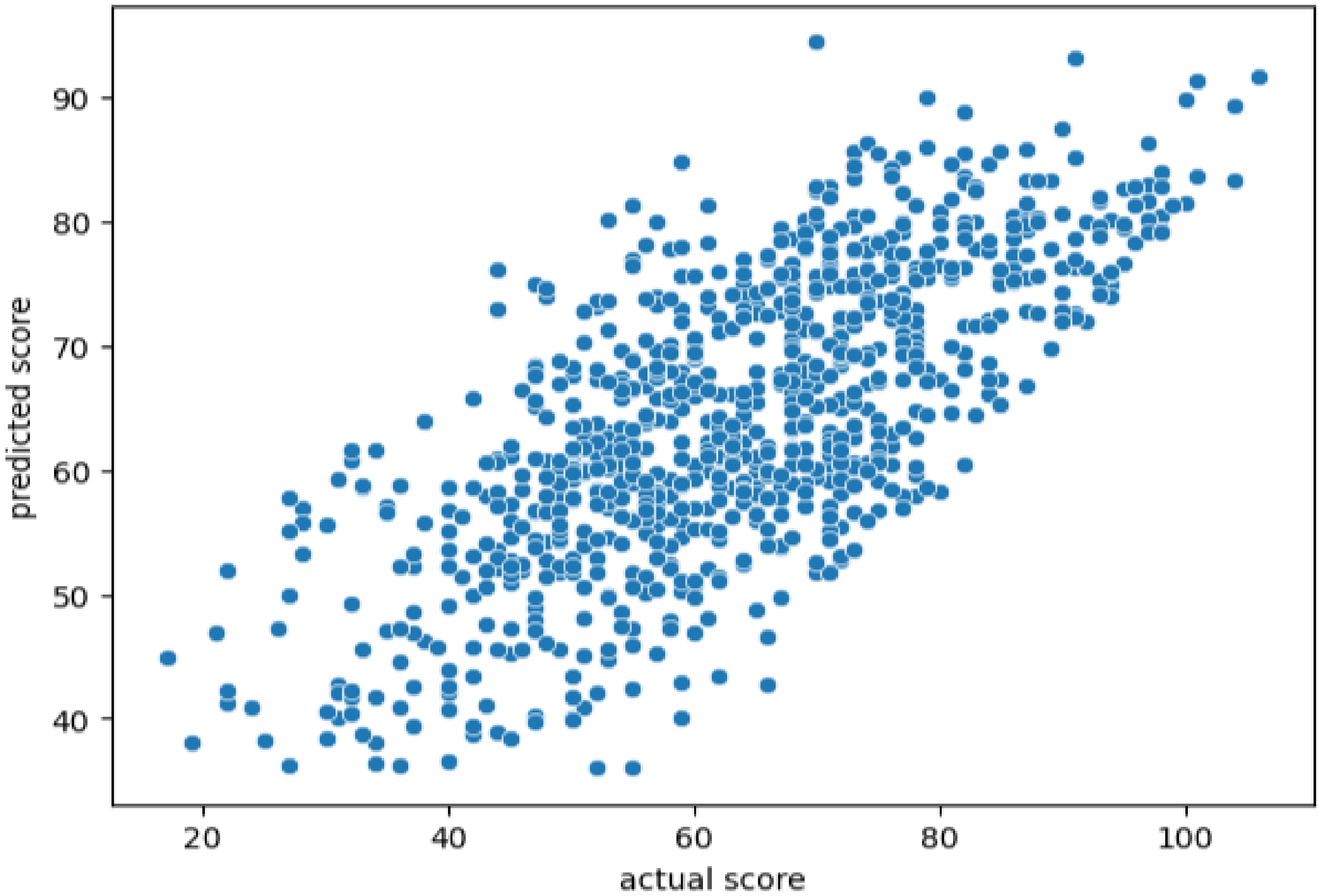
	feature	VIF
0	const	28.961879
7	Marital_single	1.521424
2	Expenses	1.44862
6	Loan_Amount	1.351979
5	Loan_Term	1.323556
10	Home_tenant	1.244445
3	Assets	1.21707
1	Income	1.197443
8	Job_freelance	1.138647
9	Job_parttime	1.078203
4	Debt	1.077881

Source: author’s analysis

An OLS model is fitted with Credit Score as the dependent variable. The model gives R^2 = 0.538 in training and R^2 = 0.514 on the test set, with $RMSE$ = 11.193 and MAE = 9.162 (see Figure 4). Income and assets raise credit score, while debt, loan term, loan amount, being single, freelance, part-time or a tenant reduce it. This makes the final model interpretable and useful.

Figure A4

Scatterplot for actual and predicted sales amount



Source: author's analysis

PART B

Abstract

This report develops and evaluates a bankruptcy prediction model using the dataset provided for the coursework. Because the target variable is highly imbalanced, the analysis focuses not only on accuracy but also on recall, F1-score, ROC-AUC and average precisions. A regularised logistic regression model is used as the benchmark, while XGBoost is applied as the advanced classifier. Both models are trained and tuned using stratified validation procedures to ensure a fair comparison. The results show that XGBoost outperforms logistic regression on the most important minority-class metrics, while maintaining the same overall accuracy. Additional interpretation using coefficients, feature importance and SHAP indicates that profitability, leverage, liquidity support and earnings persistence are the main drivers of bankruptcy risk. Overall, the findings suggest that XGBoost is the most suitable model for practical bankruptcy screening and early risk identification.

1. Introduction and business objective

The purpose of this project is to build a classification model that predicts whether a firm is likely to become bankrupt. According to the assignment brief, the organisation wants a model that can support more informed business decisions, while Ms. Emma Simms, manager, also expects a logistic regression model for comparison and a review of more advanced algorithms with stronger predictive performance. In addition, the report must identify which factors matter most and explain how they affect bankruptcy risk.

This is an important business problem because bankruptcy is a low-frequency but high-impact event. A missed bankruptcy case may lead to poor lending decisions, weak supplier selection or flawed investment screening. For that reason, the analysis should not optimise only for headline accuracy. In a strongly imbalanced dataset, a naïve model can achieve very high accuracy simply by predicting that no firm will fail. This would be operationally useless. Precision, recall, F1-score, ROC-AUC, balanced accuracy and average precision therefore provide a more informative basis for comparison.

Therefore, the project aims to deliver a reliable bankruptcy classifier that outperforms a simple benchmark and to identify the main financial signals associated with failure.

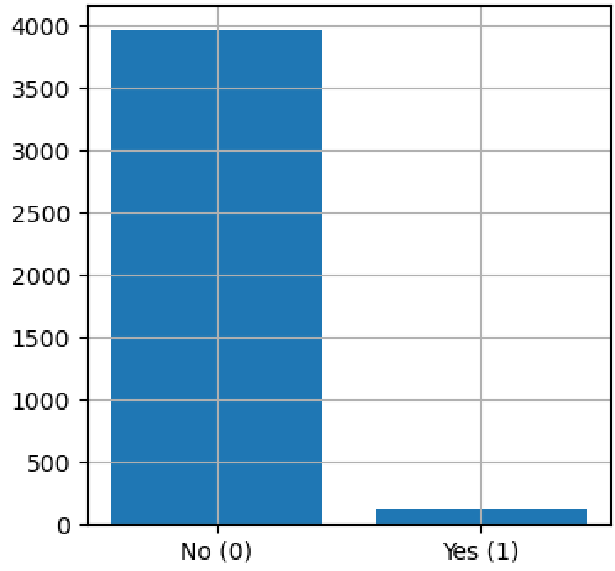
2. EDA

The exploratory stage shows that the data quality is generally good. The notebook reports no missing values and no duplicate rows. Most variables are numeric, which is appropriate for the planned modelling methods. One constant feature, `Net Income Flag`, contains only one value and is removed because such a variable carries no discriminatory information. After this step, the modelling dataset contains 94 predictors.

The key EDA findings is strong class imbalance: 96.92% non-bankrupt (0) versus 3.08% (1) bankrupt firms, so minority-class metrics are more informative than accuracy alone. Several predictor pairs are highly correlated (see Table AP1 in Appendix). This may affect coefficient stability in logistic regression, but it is less problematic for chosen advanced model and is partly controlled through logistic regularisation.

Figure B1

Class imbalance bar chart



Source: author's analysis

The notebook also examines simple correlations between each feature and the target. The strongest absolute correlations include `Net Income to Total Assets`, `ROA(A) before interest and % after tax`, `ROA(B) before interest and depreciation after tax`, `ROA(C) before interest and depreciation before interest` and `Net worth/Assets`. This already suggests that profitability and capital strength are central to bankruptcy risk. In practical terms, weaker returns on assets and weaker net worth positions are consistent with financial distress.

3. Model development

3.1 Modelling strategy

The modelling stage was designed around the business reality of bankruptcy screening rather than around raw accuracy alone. Because only about 3.08% of firms in the sample are bankrupt, a naïve classifier can obtain a very high accuracy score by always predicting the majority class, while still failing to detect distressed firms. For than reason, the notebook first establishes a simply dummy benchmark and then compares two substantive models: a regularised logistic regression and **XGBoost** classifier.

This choice creates a balanced technical design. Logistic regression is required by the brief, widely used as a transparent benchmark and allows direct interpretation of coefficient sign and odds ratios. **XGBoost** is selected as the advanced model because gradient-boosted trees can capture non-linearities and variable interactions that a linear model cannot represent directly (Chen and Guestin, 2016).

The dataset was split into training, validation and test subsets using stratified sampling. The first split held out 20% of observations for final testing, while the remaining 80% were divided into 75 % training and 25% validation. This produces an effective 60/20/20 structure and preserves the bankruptcy proportion in every subset. Therefore, the modelling logic is sequential. Train candidate models on the training set, tune hyperparameters using cross-validation, compare competing models on the validation set, optimise the decision threshold for the preferred business objective and evaluate only once on the untouched test set. This structure reduces information leakage and makes the final performance estimate more credible. For the logistic model, the notebook uses a pipeline with median imputation, standardisation and a regularised **LogisticRegression** estimator. The probability is modelled as:

$$P\ (Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)}}$$

Here, $Y = 1$ indicates bankruptcy, $x_1, \dots, x_k$ are the financial predictors, β_0 is the intercept and $\beta_1, \dots, \beta_k$ are coefficients learned from the data. Because the classes are imbalanced, `class_weight = “balanced”` is applied and the solver is set to **saga**, so that both L1 and L2 penalties can be tested. **RandomizedSearchCV** is used to search over the regularisation strength C and the penalty type. This is sensible design because regularisation shrinks unstable coefficients, reduces overfitting and can implicitly perform feature selection when the L1 penalty is chosen (Hastie, Tibshirani and Friedman, 2009; scikit-learn, 2026a).

For the advanced model, the notebook fits **XGBoost** with `objective = “binary:logistic”` and `eval_metric = “logloss”`. The search space covers the number of trees, learning rate, tree depth, subsampling, column subsampling, minimum child weight and L1/L2 regularisation terms. In addition, `scale_pos_weight` is set from the class ratio in the training data, so that the algorithm penalises misclassified bankrupt firms more strongly. This is an important business adjustment because the cost of overlooking a bankrupt company is typically much larger than the cost of reviewing a false positive. Five-fold stratified cross validation is used in both model searches and the search objective is Average Precision rather than accuracy. That is methodologically appropriate for rare-event classification because the precision-recall framework is more informative than accuracy and often more informative than ROC curves, when the positive class is uncommon (Saito and Rehmsmeier, 2015).

3.2 Initial validation comparison

The dummy benchmark confirms why accuracy cannot be treated as the main decision metric in this case. On the validation set, the most frequent classifier achieved 96.94% accuracy, but recall and F1 for the bankrupt class were both zero. In business terms, the model looks successful numerically, while being operationally useless. This justifies the use of F1, recall, balanced accuracy, ROC-AUC and Average Precision in the remainder of the report.

Table B1

Comparison of models on validation datasets

Model	Accuracy	Balanced_ Accuracy	Precision	Recall	F1	ROC_AUC	Avg Precision
Logistic Regression	0.8655	0.8725	0.1705	0.88	0.2857	0.929	0.3506
XGBoost	0.9352	0.7148	0.2308	0.48	0.3117	0.9112	0.3334

Source: author's analysis

Table B1 shows that 2 models behave very differently at the default 0.50 threshold. Logistic regression is much more aggressive. It captures 88% of bankrupt firms but with low precision and much lower overall accuracy. **XGBoost** is more conservative. It delivers a slightly better F1 score, materially higher precision and higher accuracy, but lower recall. At this stage neither model should be selected using the default threshold alone. The next step is to align the threshold with the business objective (see Figure AP1). Threshold tuning materially improves both classifiers. The classification rule becomes:

$$\hat{y} = 1 \; if \; p \geq \tau, otherwise \; 0$$

Rather than fixing τ at 0.50, the notebook searches over candidate thresholds and selects the value that maximises the F1 score on the validation set. For **logistic regression**, the best threshold is 0.91, for **XGBoost**, it is 0.72. The very high optimal threshold for logistic regression suggests that the model produces a wide range of positive probabilities and needs strong filtering to reduce false alarms **XGBoost** also benefits from threshold adjustment but remains less extreme.

3.3 Final test evaluation

After threshold selection, both tuned models were refitted on the combined training and validation data and then evaluated on the test set. This is the most important comparison in the report because it approximates out-of-sample performance under a realistic deployment workflow. The results are summarised below (see Table B2).

Table B2

Refitted models comparison

Model	Threshold	Accuracy	Balanced_ Accuracy	Precision	Recall	F1	ROC_ AUC	Avg Precision
XGBoost	0.72	0.9634	0.7487	0.4194	0.52	0.4643	0.9468	0.4786
Logistic Regression	0.91	0.9634	0.6906	0.4	0.4	0.4	0.9295	0.4401

Source: author's analysis

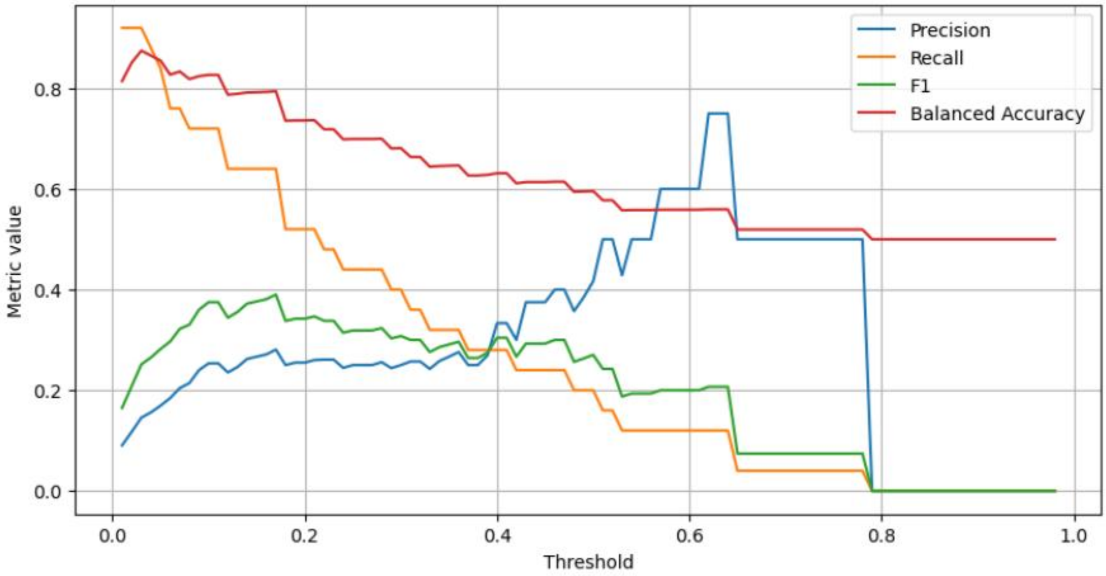
XGBoost is the strongest final classifier. Both models achieve the same overall accuracy of 96.34%, which shows why accuracy is not discriminating enough in this project. Once the focus shifts to minority-class detection, **XGBoost** is clearly superior. It raises recall from 0.40 to 0.52, improves precision from 0.40 to 0.419 and increases F1 from 0.400 to 0.464. Balanced accuracy also improves from 0.691 to 0.749. In operational terms, the advanced model identifies more bankrupt firms while preserving a similar false-positive burden. For a screening application, that is a meaningful improvement. Figure AP2 represents the confusion matrices for the tuned models. Both models correctly classify most non-bankrupt firms, while **XGBoost** detects a larger share of bankrupt companies, confirming its stronger performance in identifying financially distressed firms.

3.4 Probability calibration

A further enhancement tested in the notebook is probability calibration for the final **XGBoost** model using isotonic regression through **CalibratedClassifierCV**. This step is highly relevant from a business perspective. A ranking model may be acceptable for triage, but many practical use cases require probabilities that can be interpreted as calibrated risk levels for escalation rules, manual review threshold or portfolio monitoring. Tree based ensembles often produce well-ranked but imperfectly calibrated probabilities, so recalibration is a justified extension (Niculescu-Mizil and Caruana, 2005; scikit-learn, 2026c).

Figure B2

Threshold tuning for calibrated **XGBoost** on the validation set



Source: author's analysis

Calibration changes the model profile rather than dominating it across every metric (see Table B3). The calibrated **XGBoost** model achieves the highest ROC-AUC (0.9532), the highest Average Precision (0.5242) and the highest recall (0.56), but its precision falls to 0.333 and its F1 scope drops below the uncalibrated **XGBoost** model. Therefore, the choice between the 2 **XGBoost** versions depends on the organisation's decision rule. If the organisation prioritises maximising the capture of potential bankruptcies for follow-up review, the calibrated model is attractive. If it wants the best overall balance between misses and false alarms under one threshold, the uncalibrated **XGBoost** is preferable. On balance, the normal **XGBoost** model is selected as the main recommendation because F1 is highest and the business brief asks for a model that can be trusted and used rather than only a probability-ranking engine.

Table B3

Final comparison of all three models

Model	**Threshold**	**Accuracy**	**Balanced_ Accuracy**	**Precision**	**Recall**	**F1**	**ROC _AUC**	**Average _Precision**
XGBoost	0.72	0.9634	0.7487	0.4194	0.52	0.4643	0.9468	0.4786
Calibrated XGBoost	0.17	0.9524	0.7624	0.3333	0.56	0.4179	0.9532	0.5242
Logistic Regression	0.91	0.9634	0.6906	0.4	0.4	0.4	0.9295	0.4401

Source: author's analysis

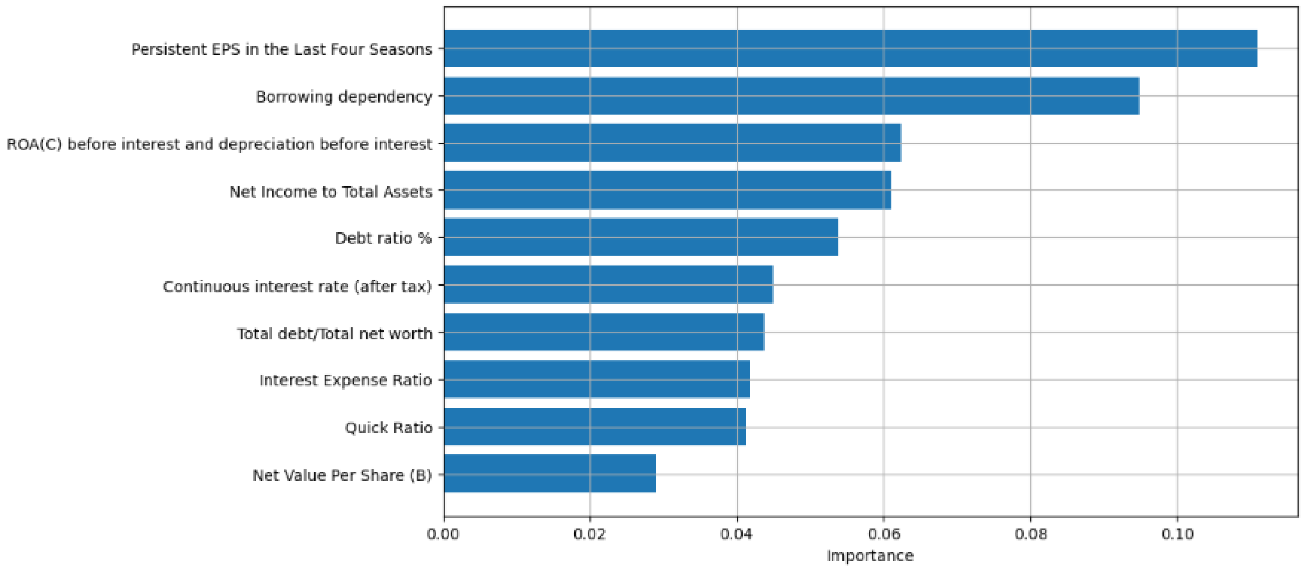
5. Model interpretation and insights

For logistic regression, coefficient signs show the direction of association after standardisation and the odds ratio e^{β_t} shows how one-standard-deviation increase in a feature changes the odds of bankruptcy (other variables are constant). The largest positive coefficients include `Debt Ratio %`, `Current Asset Turnover Rate`, `Revenue per person` and `Current Liability to Assets`. The largest negative coefficients include `Total Asset Turnover`, `Persistent EPS in the Last Four Seasons`, `ROA(C) before interest and depreciation before interest`, `Net worth/Assets` and `Cash/Total Assets` (see Table AP2 in Appendix). Message is intuitive: leverage and balance-sheet pressure increase bankruptcy risk, while profitability, earnings persistence, liquidity and asset efficiency reduce it. This pattern is consistent with classic bankruptcy prediction literature starting from Altman (1968) and later formulations such as Ohlson (1980).

The XGBoost importance plot refines this view (see Figure B3). This is useful because tree-based importance captures non-linear split contributions, not just linear marginal effects.

Figure B3

Top 10 XGBoost feature importances



Source: author's analysis

SHAP analysis strengthens the interpretation because it quantifies both global importance and the directions of local impact for the final model (Lundberg and Lee, 2017). The beeswarm plot indicates that lower profitability, weaker operating funds relative to liabilities and lower earnings persistence push predictions toward the bankrupt class (see Figure AP2 in Appendix). This is exactly the type of insight that the client can use beyond pure prediction. The model does not merely classify firms: it's highlighting a coherent financial-risk narrative centred on profitability erosion, debt dependence and reduced internal funding capacity.

6. Limitations and recommendations

The organisation can benefit because the selected classifier provides a practical bankruptcy-screening framework. The model can support early identification of financially distressed firm before bankruptcy is formally observed, making it useful for lending decisions, supplier due diligence, investment screening, insurance underwriting and internal counterparty monitoring. The calibrated version also provides a suitable basis for probability-based risk scoring if the organisation later requires estimated default likelihoods rather than class labels one.

Several limitations remain important. First, the dataset was provided without a full data dictionary, so some variables can only be interpreted approximately. As a result, parts of the analysis should be viewed as pattern based rather than fully causal in an economic sense. Second, substantial multicollinearity exists among several ratios. This is not a major issue for XGBoost, but it means logistic regression coefficients should be interpreted directionally rather than as isolated effects. Third, the dataset is strongly imbalanced, so model usefulness still depends on the relative cost of false negatives and false positives. Finally, the analysis relies on one historical dataset only; before deployment, the model should be validated on newer periods and monitored for concept drift.

The recommended deployment choice is that tuned XGBoost model with a threshold of 0.72 for primary screening, supported by SHAP-based monitoring of the main drivers. This option provides the best minority-class balance and the highest F1 score while remaining interpretable. Where missed bankruptcies are considered more costly than additional manual reviews, the calibrated XGBoost version is preferable because it increases recall. Therefore, outcome is not just a single model, but a flexible framework that can be aligned with different risk appetites.

Reference

Altman, E.I. (1968) 'Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy", The Journal of Finance, 23(4), pp. 589-609. Available at: https://doi.org/10.1145/2939672.2939785

Chen, T. and Guestrin, C. (2016) 'XGBoost: A Scalable Tree Boosting System' Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785-794. Available at: https://doi.org/10.1145/2939672.2939785

Hastle, T., Tibshirani, R. and Freidman. J (2009) The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd edn. New York: Springer. Available at: https://hastie.su.domains/ElemStatLearn/

Lundberg, S.M. and Lee, S.-I. (2017) 'A Unified Approach to Interpreting Model Predictions", Advances in Neural Information Processing System, 30. Available at: https://hastie.su.domains/ElemStatLearn/ https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf

Niculescu-Mizil, A. and Caruana, R. (2005) 'Predicting Good Probabilities With Supervised Learning", Proceedings of the 22nd International Conference on Machine Leaning, pp. 625-632. Available at: https://www.cs.cornell.edu/~alexn/papers/calibration.icml05.crc.rev3.pdf

Ohlson, J.A. (1980) 'Financial Ratios and the Probabilistic Prediction of Bankruptcy', Journal of Accounting Research, 18(1), pp. 109-131. Available at: https://www.jstor.org/stable/2490395

Saito, T. and Rehmsmeier, M. (2015) 'The Precision-Recall Plot is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets', PLoS ONE 10(3), e0118432. Available at: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0118432

scikit-learn(2026a) LogisticRegression. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

scikit-learn(2026b) Probability Calibration. Available at: https://scikit-learn.org/stable/modules/calibration.html

scikit-learn(2026c) CalibratedClassifierCV. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.calibration.CalibratedClassifierCV.html

Appendix

Table AP1

Predictor pairs with correlation coefficients more than 0.90

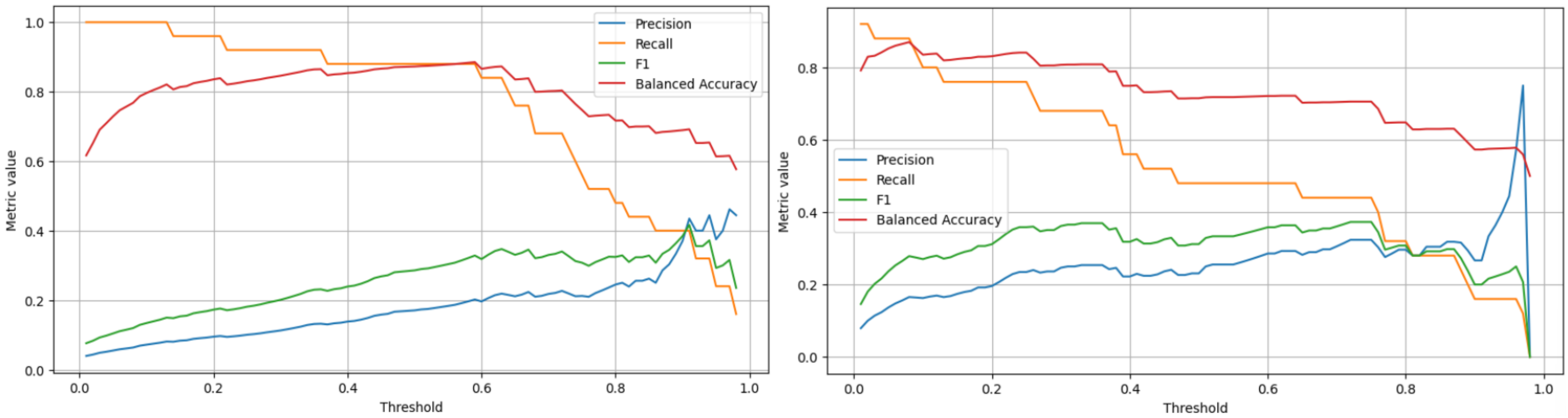
Feature_1	Feature_2	Correlation
Current Liabilities/Liability	Current Liability to Liability	1
Current Liabilities/Equity	Current Liability to Equity	1
Debt ratio %	Net worth/Assets	1
Revenue Per Share (Yuan ¥)	Total debt/Total net worth	1
Operating Gross Margin	Gross Profit to Sales	0.999999991
Net Value Per Share (A)	Net Value Per Share (C)	0.999809589
Operating Gross Margin	Realized Sales Gross Margin	0.999455745
Realized Sales Gross Margin	Gross Profit to Sales	0.999455697
Operating Profit Per Share (Yuan ¥)	Operating profit/Paid-in capital	0.999304944
Pre-tax net Interest Rate	Continuous interest rate (after tax)	0.999267055
After-tax Net Profit Growth Rate	Regular Net Profit Growth Rate	0.999202816
Net Value Per Share (B)	Net Value Per Share (A)	0.998972518
Net Value Per Share (B)	Net Value Per Share (C)	0.998781229
After-tax net Interest Rate	Continuous interest rate (after tax)	0.98873697
Pre-tax net Interest Rate	After-tax net Interest Rate	0.986121684
ROA(C) before interest and depreciation before interest	ROA(B) before interest and depreciation after tax	0.98602935
Working capitcal Turnover Rate	Cash Flow to Sales	0.983048219
Current Liabilities/Equity	Liability to Equity	0.975413432
Current Liability to Equity	Liability to Equity	0.975413432
Persistent EPS in the Last Four Seasons	Net profit before tax/Paid-in capital	0.971571034
Per Share Net profit before tax (Yuan ¥)	Net profit before tax/Paid-in capital	0.971452492
ROA(A) before interest and % after tax	Net Income to Total Assets	0.955749903
Operating Profit Rate	Pre-tax net Interest Rate	0.953308377
Borrowing dependency	Liability to Equity	0.951335221
ROA(A) before interest and % after tax	ROA(B) before interest and depreciation after tax	0.951183896

Persistent EPS in the Last Four Seasons	Per Share Net profit before tax (Yuan ¥)	0.950556133
Operating Profit Rate	Continuous interest rate (after tax)	0.947485873
ROA(C) before interest and depreciation before interest	ROA(A) before interest and % after tax	0.934393116
Cash flow rate	Operating Funds to Liability	0.929663609
Borrowing dependency	Current Liabilities/Equity	0.907615816
Borrowing dependency	Current Liability to Equity	0.907615816
ROA(B) before interest and depreciation after tax	Net Income to Total Assets	0.90256738

Source: author's analysis

Figure AP1

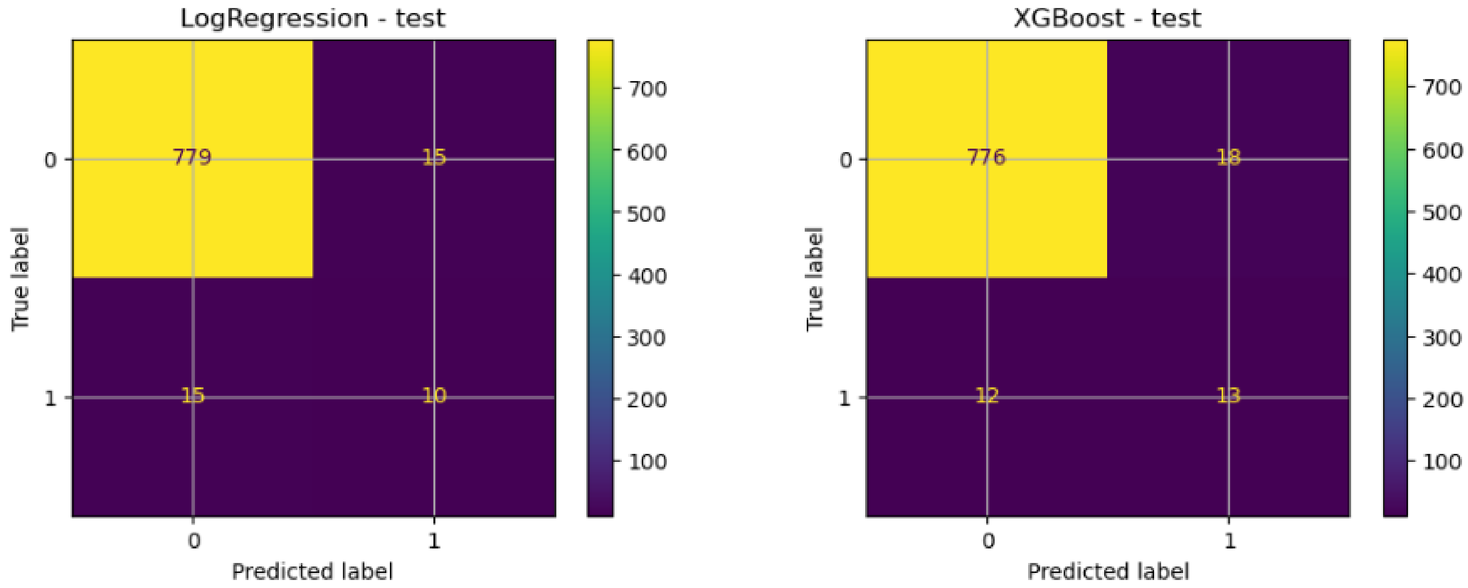
Threshold tuning for logistic regression (to the left) and **XGBoost** (to the right) on validation dataset



Source: author's analysis

Figure AP2

Test confusion matrices for the tuned models



Source: author's analysis

Table AP2

Interpretation of logistic regression odds ratios

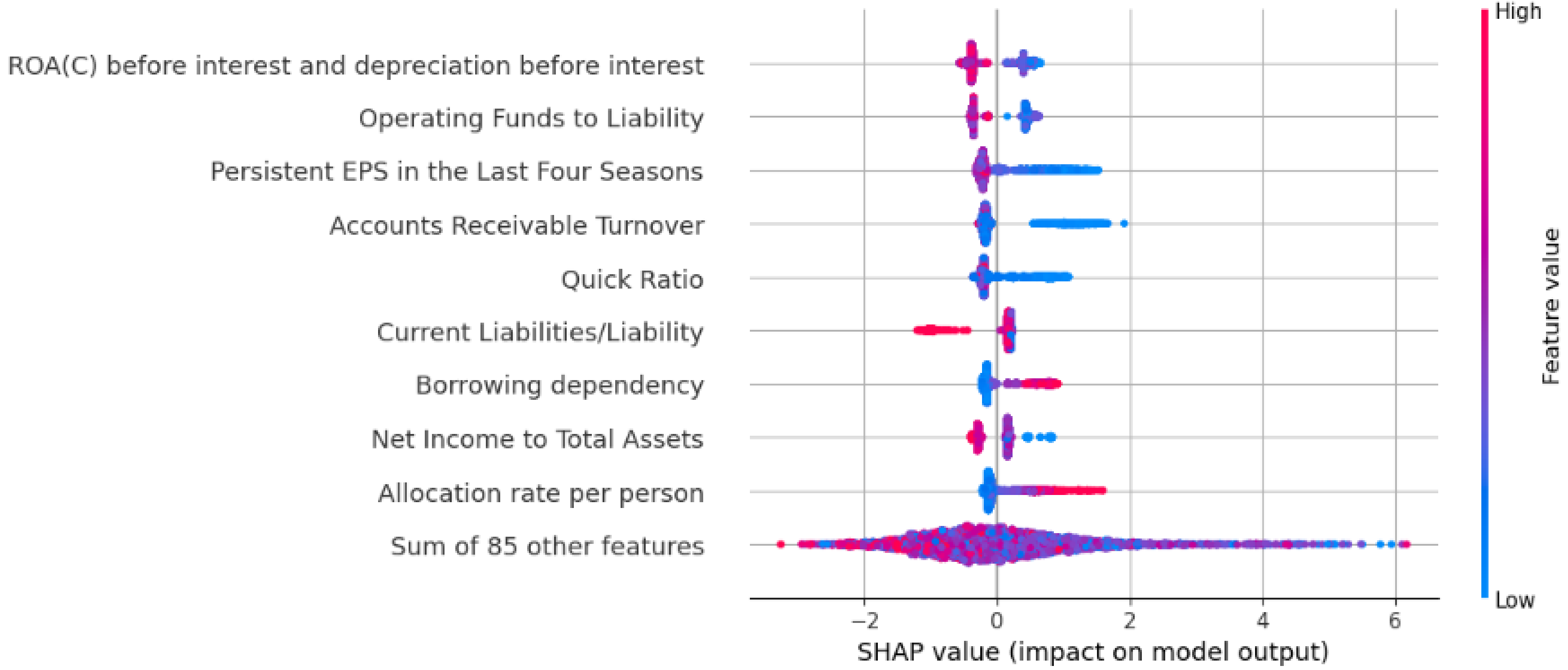
Feature	**Direction**	**Odds ratio**	**Interpretation**
`Debt Ratio %`	+	2.017	Higher leverage raises bankruptcy odds materially
`Current Asset Turnover Rate`	+	1.318	Higher working capital pressure is associated with distress
`Current Liability to Assets`	+	1.138	Short term obligations intensify risk
`Total Asset Turnover`	-	0.353	Operational efficiency reduces bankruptcy odds

Persistent EPS in the Last Four Seasons	-	0.465	Stable earnings are protective
Cash/Total Assets	-	0.501	Liquidity buffers lower distress risk

Source: author's analysis

Figure AP3

SHAP beeswarm plot for the **XGBoost** model



Source: author's analysis